2026 NISR Infographic Competition

Mathematics for Machine Learning, Cohort 3 | African Leadership University

Team members

1. IZA Prince Boaz (team leader) - b.iza@alustudent.com
2. Bior Aguer Kuir - b.aguerkui@alustudent.com

NISR team name

We could not remember the team name we registered with, so we submitted without selecting it. We emailed the NISR organising team on 31 October 2026 asking them to link our entry to the team leader's registration email (b.iza@alustudent.com).

Project

The Invisible Half: Rwanda Vital Statistics 2019-2025

Our entry looks at birth and death registration in Rwanda using the NISR Vital Statistics Report 2025 tables. In 2025, 92.9% of births were registered but only 50.5% of deaths, so about 38,521 deaths were not recorded. We fitted a straight line with the normal equation and a logistic curve with gradient descent to estimate when death registration could reach 90% (around 2031 to 2033 if the 2021-2025 trend continues). We tested both models on 2024 and 2025, and used probability to compare men (57.6% of deaths registered) and women (43.7%).

Submitted to NISR

Date and time: 31 October 2026, 8:28 PM

Static entry: poster_invisible_half.png (poster, attached as page 3)

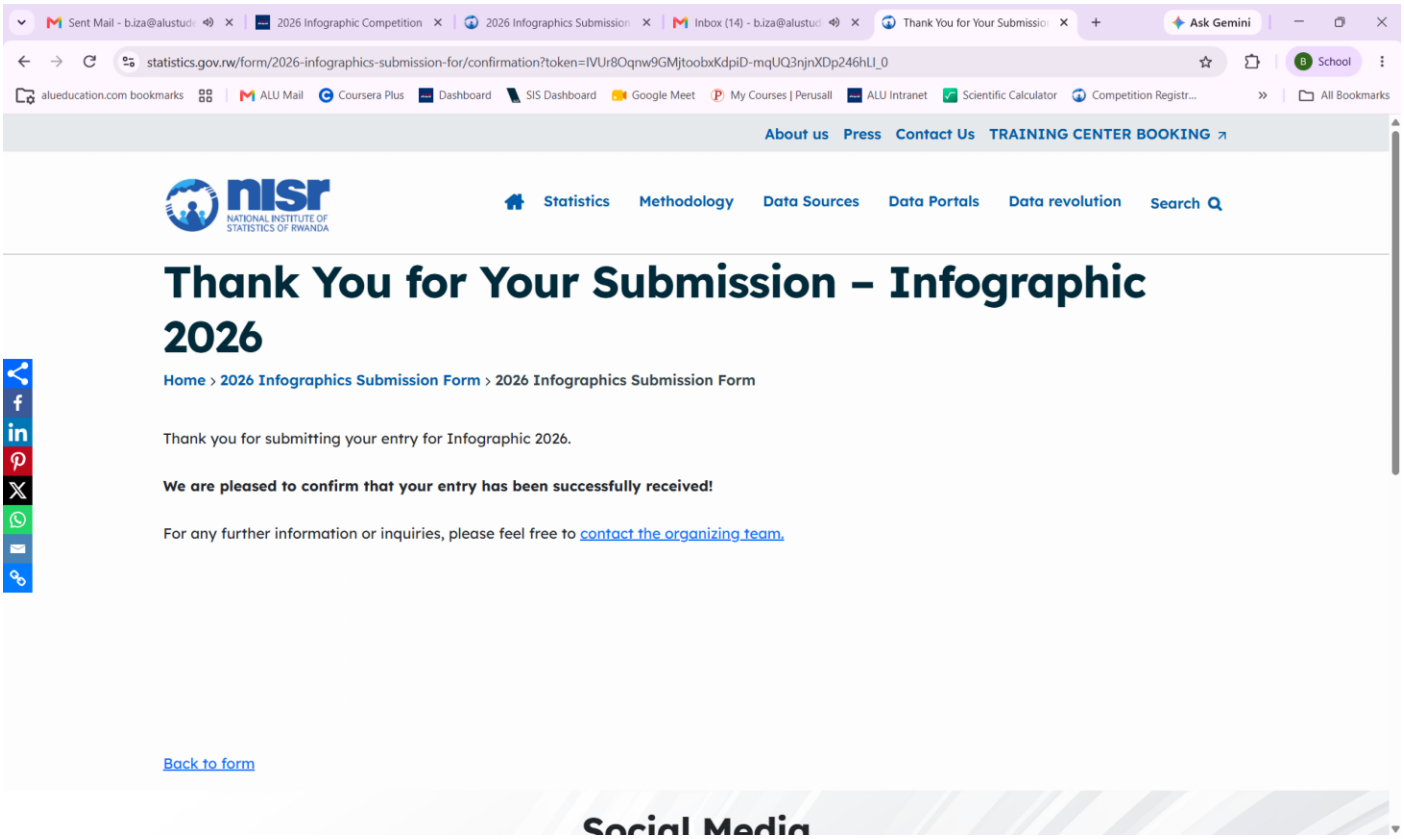
Dynamic entry: dynamic_invisible_half.mp4 (41-second video, key frames on page 4)

Proof of submission: NISR confirmation page (page 2)

Data source

National Institute of Statistics of Rwanda (NISR), Rwanda Vital Statistics Report 2025 and Final Tables_CRVS_2025.xlsx (CRVS system, 2025).

Proof of submission: NISR confirmation page



Static entry submitted to NISR

THE INVISIBLE HALF

Rwanda records almost every birth, but only half of its deaths · Vital statistics 2019–2025

92.9%

of births were recorded in 2025

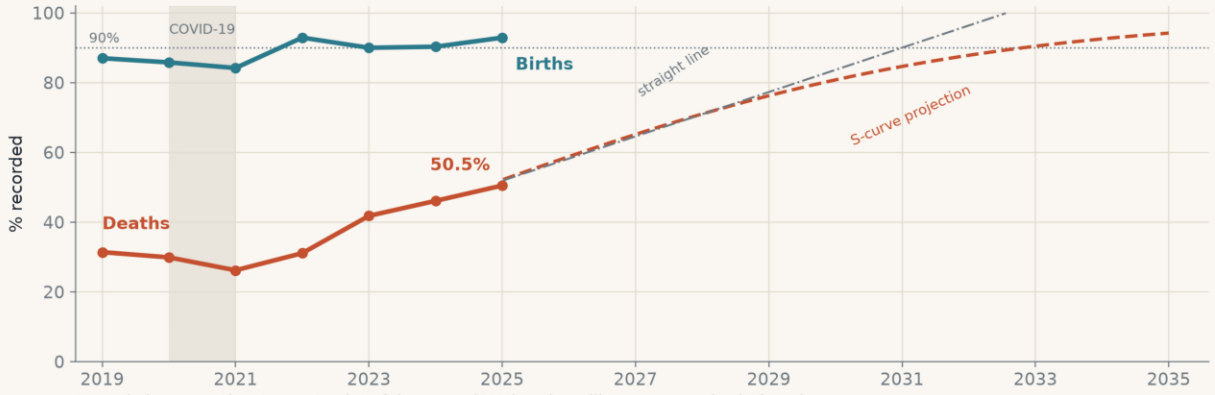
50.5%

of deaths were recorded in 2025

38,521 deaths went unrecorded in 2025: about 106 every day.

1 The gap is closing, fast

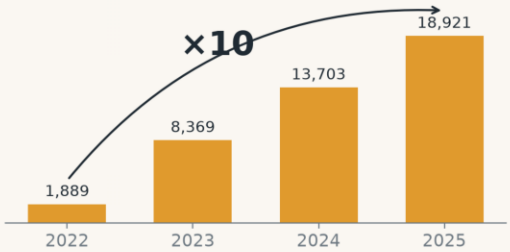
Deaths recorded rose from 26% (2021) to 50.5% (2025). If this pace continues, 90% would be recorded by about 2031 (straight line) to 2033 (S-curve, fitted by gradient descent).



Rwanda is now at the steepest point of the curve (50%): gains will come more slowly from here.

2 What changed?

Verbal autopsies (family interviews about deaths outside health facilities) grew tenfold.



This coincided with the 2022–25 rise in recorded deaths. Old deaths are being registered too: 328 → 14,517 deaths registered over a year late (2022 → 2025).

3 Women's deaths are counted less

Out of every 10 deaths in 2025, recorded:

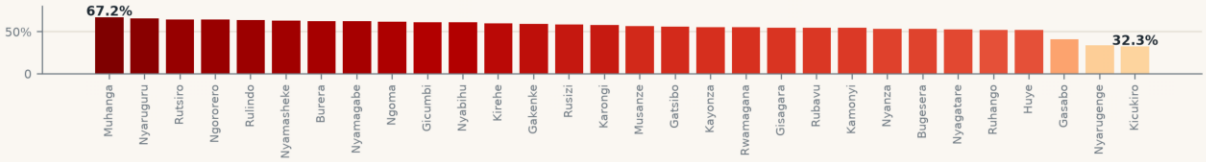
Men 57.6%

Women 43.7%

Men are 49% of all deaths but 56% of recorded ones, so statistics from registered deaths over-represent men.

4 Where deaths happen outside health facilities

Share of registered deaths in the community, 2025. Rural 61% vs urban 41%. These are the hardest deaths to record.



Source: NISR, Rwanda Vital Statistics Report 2025 (CRVS), Final Tables_CRVS_2025.xlsx. Where tables differ, Table 12 values are used (female 2025 = 43.7%, from Table 12 counts). Projections assume the 2021–25 trend continues (7 data points); 'coincided' does not mean 'caused'. Community = outside a health facility.

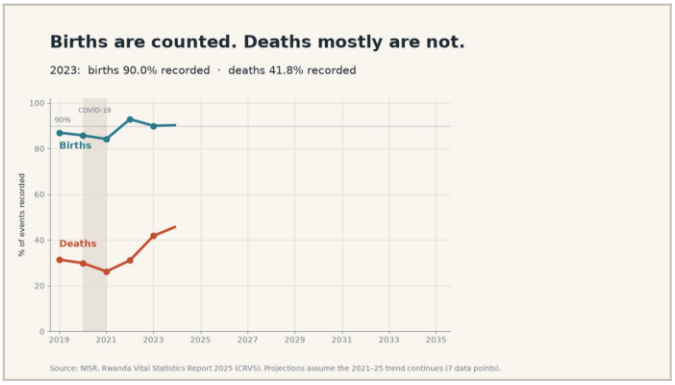
Policy: expand verbal autopsies and community death reporting where most deaths happen outside facilities, and close the gap for women.

Dynamic entry submitted to NISR: key frames

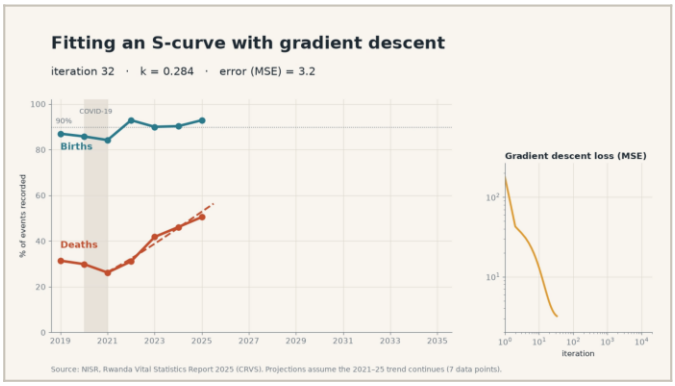
dynamic_invisible_half.mp4 | 41-second video, 1280 x 720 | full video uploaded separately



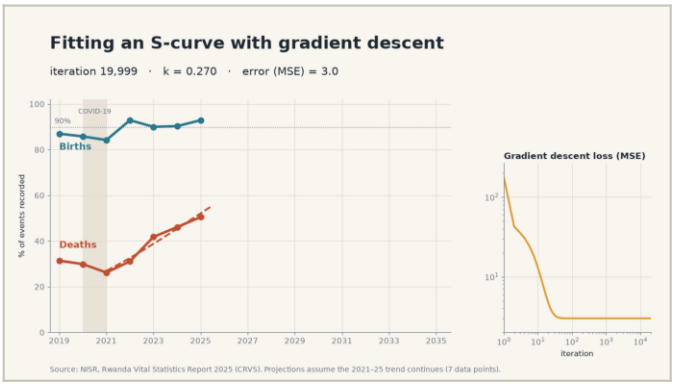
0:02 Title



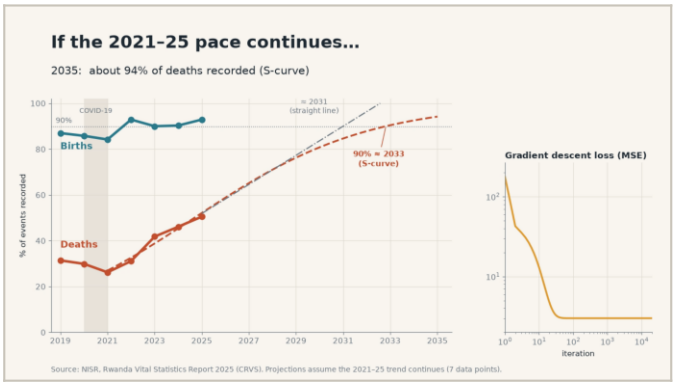
0:10 Data appears year by year



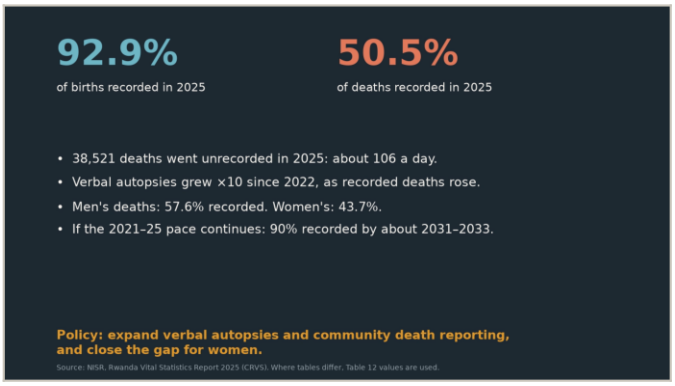
0:16 Gradient descent fits the S-curve



0:24 Fit finished, loss falls to 3.0



0:34 Projection: 90% around 2031-2033



0:39 Key messages